



UNIVERSITEIT VAN AMSTERDAM

Asymptotic Statistics

NOTES BY

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INTRODUCTION. The course starts with a review of various concepts of stochastic convergence (e.g. convergence in probability or in distribution) and properties of the multivariate normal distribution. Then the asymptotic properties of different statistical procedures are studied, including χ^2 -tests, Moment estimators, and M -estimators (including maximum likelihood estimations).

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Stochastic Convergence

SECTION 1.1

| Basic Theory

DEFINITION 1.1 (Random Variable).

A *random variable* is a measurable map $X : \Omega \rightarrow \mathcal{X}$ from a probability space $(\Omega, \mathcal{F}, \mathcal{P})$ into some measurable space $(\mathcal{X}, \mathcal{B})$.

Remark. $\mathcal{X} = \mathbb{R}$ denotes the real random variable, and $(\mathcal{X}, d)$ denotes metric spaces.

DEFINITION 1.2.

We say that a sequence of random variables (X_n) converges to a random variable X *almost surely* if

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} d(X_n, X) = 0\right) = 1.$$

And (X_n) is said to *converges in probability* to X if

$$\lim_{n \rightarrow \infty} \mathbb{P}(d(X_n, X) > \varepsilon) = 0, \quad \forall \varepsilon > 0.$$

EXAMPLE 1.3 (Law of Large Numbers). Let $X_1, X_2, \dots$ be a sequence of *independent and identically distributed (i.i.d.)* random variables according to a single observation P_0 . Define the probability space $\Omega = \mathbb{R}^\infty, \mathcal{F} = \mathcal{B}^\infty, \mathcal{P} = P_0^\infty$, and assume that X is P_0 -integrable. The (strong) law of large numbers says, if the expectation $P_0|X| < \infty$, then the sample average

$$\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$$

converges *almost surely* to P_0X (a constant random variable) with probability one.

EXAMPLE 1.4 (Central Limit Theorem). In the previous example, assume that $P_0(X^2) < \infty$. Then $\forall x \in \mathbb{R}$, the *central limit theorem* says

$$P\left(\sqrt{n} \frac{\bar{X}_n - P_0X}{\sqrt{\text{Var}(X)}} \leq x\right) \rightarrow \Phi_{0,1}(x).$$

This is called the *weak convergence* of

$$Z_n = \sqrt{n} \frac{\bar{X}_n - P_0X}{\sqrt{\text{Var}(X)}} \xrightarrow{P_0} Z \sim \mathcal{N}(0, 1).$$

DEFINITION 1.5 (Weak Convergence).

A sequence of random vectors X_n is said to *converge in distribution* (weak convergence/convergence in law) to a random vector X if

$$\mathbb{P}(X_n \leq x) \rightarrow \mathbb{P}(X \leq x)$$

for every x at which the distribution function $x \rightarrow \mathbb{P}(X \leq x)$ is continuous.

LEMMA 1.6 (Portmanteau Lemma).

Let (X_n) be a sequence of random variables and let X be a random variable, we have:

- (ii) $Pf(X_n) \rightarrow Pf(X)$ for all bounded continuous $f : \mathcal{X} \rightarrow \mathbb{R}$.
- (iii) $Pf(X_n) \rightarrow Pf(X)$ for all bounded Lipschitz $f : \mathcal{X} \rightarrow \mathbb{R}$.
- (iv) $\liminf_{n \rightarrow \infty} Pf(X_n) \geq Pf(X)$ for all non-negative continuous $f : \mathcal{X} \rightarrow \mathbb{R}$.
- (v) $\liminf_{n \rightarrow \infty} P(X_n \in G) \geq P(X \in G)$ for all open $G \subset \mathcal{X}$.
- (vi) $\limsup_{n \rightarrow \infty} P(X_n \in F) \leq P(X \in F)$ for all closed $F \subset \mathcal{X}$.
- (vii) $\lim_{n \rightarrow \infty} P(X_n \in B) = P(X \in B)$ for all Borel sets $B \subset \mathcal{X}$ such that $P(X \in \partial B) = 0$.
- (i) In the special case $\mathcal{X} = \mathbb{R}$, $P(X_n \leq x) \rightarrow P(X \leq x)$ for all $x \in \mathbb{R}$ such that $y \mapsto P(X \leq y)$ is continuous at x .

Proof. (ii) $\rightarrow$ (iii) Every f that is bounded and Lipschitz is bounded continuous.

(iii) $\rightarrow$ (v) Let G open in $\mathcal{X}$ be given, define $\forall m \geq 1$ we have

$$f_m(x) = \min\{1, md(x, G^c)\}, \quad \forall x \in \mathcal{X}.$$

□

EXAMPLE 1.7. Let $X_1, X_2, \dots$ be i.i.d. from $U[0, 1]$. Define $Y_n = \max\{X_1, \dots, X_n\}$ where $n \geq 1$. Calculate that

$$P(-n(Y_n - 1) > x) = \left(1 - \frac{x}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-x}.$$

The random variable $Y \sim \exp(1)$, $Y_n \xrightarrow{P} Y$ by Portmanteau (i).

THEOREM 1.8 (Continuous Mapping Theorem).

If $X_n \rightarrow X$ and $g : x \rightarrow y$ continuous, then $g(X_n) \rightarrow g(X)$.

Proof. $Pf(Z_n) \rightarrow Pf(Z)$ for all bounded continuous $f : y \rightarrow \mathbb{R}$. Then $P(f \circ g)(X_n) \rightarrow P(f \circ g)(X)$ because $x \rightarrow y$ and $y \rightarrow \mathbb{R}$ are continuous. □

PROPOSITION 1.9.

Any CDF $x \mapsto P(X \leq x)$ has at most a countable number of discontinuities.

Proof. Let $F(x) = P(X \leq x) \in [0, 1]$, define

$$D_n = \left\{x \in \mathbb{R} : F(x) - \lim_{y \uparrow x} F(y) > \frac{1}{n}\right\}, \quad n \geq 1.$$

The set of all discontinuities is $D = \bigcup_{n \geq 1} D_n$, then $\#(D_n) < n + 1$. If $n + 1$ jumps of size $\frac{1}{n}$, $F(x) > 1$ for some x . Therefore $\#(D)$ countable. □

DEFINITION 1.10.

Any random variable in $\mathbb{R}$ (or $\mathbb{R}^n$) is *tight*:

$$P(\|X\| < \infty) = 1.$$

Equivalently, $\forall \varepsilon > 0, \exists M > 0 : P(|X| > M) < \varepsilon$, also for every (M_n) such that

$$P(|X| > M_n) \rightarrow 0, \quad M_n \rightarrow \infty.$$

DEFINITION 1.11.

A collection $\{X_\alpha : \alpha \in I\}$ of random variables is called *uniform tight* if $\forall \varepsilon > 0, \exists M > 0$:

$$\sup_{\alpha \in I} P(\|X_\alpha\| < \varepsilon).$$

THEOREM 1.12 (Heine-Borel Theorem).

For any $A \subset \mathbb{R}$, A is compact if and only if A is bounded and closed.

COROLLARY 1.13.

Any bounded subset of $\mathbb{R}$ is relatively compact.

COROLLARY 1.14.

Any sequence inside a bounded set of $\mathbb{R}$ has a convergent subsequence.

THEOREM 1.15 (Prohorov's Theorem).

Let (X_n) be a sequence of random variables in $\mathbb{R}$.

- (i) If $X_n \rightsquigarrow X$, then $\{X_n : n \geq 1\}$ is uniformly tight.
- (ii) If X_n is uniformly tight, then (X_n) has a weakly convergent subsequence.

EXAMPLE 1.16. A sequence (X_n) that is not uniformly tight is $X_n \sim U[n, n+1]$ for $n \geq 1$.

DEFINITION 1.17.

A function $G : \mathbb{R} \rightarrow [0, 1]$ is called a *defective distribution function* if G is monotone increasing and right-continuous.

LEMMA 1.18 (Helly's Lemma).

For any sequence of defective distribution functions (F_n) , there exists a subsequence that converges to a (possibly) defective distribution function.

Remark. This characterise the counter examples as those in which the limit is defective rather than a proper defective distribution function.

Proof.

□

We may use Helly's Lemma to prove Prohorov's Theorem.

THEOREM 1.19.

Let $(X_n), (Y_n)$ be sequences of random variables and let X, Y be random variables, we have:

- (i) $X_n \xrightarrow{a.s.} X$ implies $X_n \xrightarrow{\mathcal{P}} X$.

- (ii) $X_n \xrightarrow{\mathcal{P}} X$ implies $X_n \rightsquigarrow X$.
- (iii) $X_n \rightsquigarrow c \Leftrightarrow X_n \xrightarrow{\mathcal{P}} c$ for a constant c .
- (iv) If $X_n \xrightarrow{\mathcal{P}} X$ and $d(X_n, Y_n) \xrightarrow{\mathcal{P}} 0$, then $Y_n \xrightarrow{\mathcal{P}} X$.
- (v) If $X_n \rightsquigarrow X$ and $Y_n \xrightarrow{\mathcal{P}} c$ for a constant c , then $(X_n, Y_n) \rightsquigarrow (X, c)$.
- (vi) If $X_n \xrightarrow{\mathcal{P}} X$ and $Y_n \xrightarrow{\mathcal{P}} Y$, then $(X_n, Y_n) \xrightarrow{\mathcal{P}} (X, Y)$.
- (vii) If X_n and Y_n are independent, then $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow Y$ implies $(X_n, Y_n) \rightsquigarrow (X, Y)$.

Proof. (i)

(ii)

□

LEMMA 1.20 (Slutsky).

Let X_n, X and Y_n be random vectors or variables. If $X_n \rightsquigarrow X$ and $Y_n \rightsquigarrow c$ for a constant c , then

- (i) $X_n + Y_n \rightsquigarrow X + c$.
- (ii) $X_n Y_n \rightsquigarrow cX$.
- (iii) $Y_n^{-1} X_n \rightsquigarrow c^{-1} X$ provided $c \neq 0$.

Proof. Let D_1, D_2 be subsets of $\mathbb{R}$ such that $P(X \in D_1) = 1$, and $D_1 \times \{c\} \subset D$. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous. Then by *Continuous Mapping Theorem*, $g(X_n, Y_n) \rightsquigarrow g(X, c)$. (i) $g(x, y) = x + y$. (ii) $g(x, y) = xy$. (iii) $g(x, y) = x/y$. □

SECTION 1.2

| Stochastic o and O Symbols

$$X_n = o(1) \Rightarrow x_n \rightarrow 0, \quad x_n = O(1) \Rightarrow |x_n| < M.$$

More generally, $X_n = O_P(r_n)$ with $r_n \rightarrow 0$ ($r_n > 0$) means $\frac{X_n}{r_n} = O_P(1) \Rightarrow \left| \frac{X_n}{r_n} \right| < M$.

$X_n = o_P(r_n)$ means $\frac{X_n}{r_n} = o(1) \Rightarrow \frac{X_n}{r_n} \xrightarrow{\mathcal{P}} 0$.

If a sequence of random variables (X_n) is $X_n = O_P(1)$, means that X_n is *uniformly tight*.

There are many rules of calculus with o and O symbols, for instance,

$$\begin{aligned} o_P(1) + o_P(1) &= o_P(1), \\ O_P(1) + o_P(1) &= O_P(1), \\ O_P(1)o_P(1) &= o_P(1), \\ (1 + o_P(1))^{-1} &= O_P(1). \end{aligned}$$

EXAMPLE 1.21. Let $Y_1, Y_2, \dots$ be *i.i.d.* with expectation $P|Y| < \infty$ and $PY^2 < \infty$. Let

$$T_n = \frac{\sqrt{n}\bar{Y}_n}{S_n}$$

where

$$S_n^2 = \frac{1}{n-1} (Y_i - \bar{Y}_n)^2.$$

By the *LLN*,

$$S_n^2 = \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n Y_i^2 - \bar{Y}_n^2 \right) \xrightarrow{\mathcal{P}} PY^2 - (PY)^2.$$

| Exercises

EXERCISE 1.1. Let $\mathbb{P}(X_n = \frac{i}{n}) = \frac{1}{n}$ for every $i = 1, 2, \dots, n$. Show that $X_n \rightsquigarrow X$ for a uniform variable X .

EXERCISE 1.2. Find an example of a sequence of random variables such that $X_n \rightsquigarrow 0$, but $EX_n \rightarrow \infty$.

EXERCISE 1.3. If $\sqrt{n}(T_n - \theta)$ converges in distribution, then T_n converges in probability to θ . Prove this.

Multivariate-Normal Distribution

| Covariance Matrices

SECTION 2.2

| Definition and Basic Properties

SECTION 2.3

| Multivariate Central Limit Theorem

SECTION 2.4

| Quadratic Forms

If $Z \sim \mathcal{N}(0, 1)$, then the distribution of Z^2 is called χ_1^2 -distribution (χ^2 with one degree of freedom). If $Z_1, \dots, Z_k \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$, then $Z_1^2 + Z_2^2 + \dots + Z_k^2$ has χ_k^2 -distribution.

Note that if $X \sim \mathcal{N}_k(0, \mathbb{1}_k)$, then $\|X\|^2$ has χ_k^2 -distribution. To generalise, if $X \sim \mathcal{N}_k(0, \Sigma)$, then $\|X\|^2 \stackrel{\mathcal{P}}{=} \sum_{i=1}^k \lambda_i Z_i^2$.

If X has an expectation, then $X - \mu$ has expectation zero. If $EX \neq 0$, then $\|X\|^2$ has a so-called *non-central χ^2 -distribution*.

SECTION 2.5

| χ^2 -Tests

How do we define $N_k(0, \Sigma)$ when $k = \infty$? First think of the distribution of an *i.i.d.* sample of $\Psi_1, \Psi_2, \dots$ (infinite data points), each marginally normal. We always use

$$(\Psi_1, \dots, \Psi_n) \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$$

but what about $(\Psi_i : i \geq 1)$? We have a direct answer, $N_\infty(\mu, \sigma^2)$ is a straightforward product distribution on $\mathbb{R}^\infty$. What about $\Psi_1, \Psi_2, \dots$ that forms a “vector” in $l^2 (= \{(x_n) \mid \sum x_n^2 < \infty\})$? To make a start with the answer, assume that the Ψ_i ’s are independent and $\Psi_i \sim N(0, \sigma_i^2)$.

$$\Sigma_{ij} = E[\Psi_i - E\Psi_i][\Psi_j - E\Psi_j] = E[\Psi_i\Psi_j],$$

when $i \neq j$, $\Psi_i \perp \Psi_j \Rightarrow \Sigma_{ij} = 0$.

$$E\left(\sum_{j=1}^J \Psi_j\right) = \sum_{j=1}^J \sigma_j^2$$

$$\Sigma = \begin{pmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \end{pmatrix}$$

$$\begin{aligned} Tr(\Sigma) &= \lim_{J \rightarrow \infty} \sum_{j=1}^J E[\Psi_j^2] \\ &= \lim_{J \rightarrow \infty} E \sum_{j=1}^J \Psi_j^2 \quad \text{increasing} \\ &= E \lim_{J \rightarrow \infty} \sum_{j=1}^J \Psi_j^2 = E\|\Psi\|^2 \end{aligned}$$

by monotone convergence theorem.

$$P(\|\Psi\|^2 > M^2) \leq \frac{E\|\Psi\|^2}{M^2} < \varepsilon$$

by Markov’s inequality. As $M \rightarrow \infty$, $P(\|\Psi\| > M) \rightarrow 0$ which means that $\|\Psi\| < \infty$ with probability one. If we start with dependent Ψ_i ’s, Σ_{ij} general (not 0 for $i \neq j$). $Tr(\Sigma) = E\|\Psi\|^2$ still holds with the same conclusion

$$P(\Psi \in l^2) \Leftrightarrow Tr(\Sigma) < \infty.$$

What about asymptotic aspects of estimation? For any expectation value, $E[f(X)]$ we may estimate by

$$T_n(X_1, \dots, X_n) = \frac{1}{n} \sum_{i=1}^n f(X_i)$$

if $X_1, X_2, \dots$ *i.i.d.*, $E[f(X)] < \infty$. The LLN: $T_n \xrightarrow{a.s.} E[f(X)]$. In most questions $X \sim P_{\theta_0}$ where $\theta_0 \in \Theta$ (estimator space) $\in \mathbb{R}^k$ and we want to estimate θ_0 . If $\theta = E_\theta f(X)$ then

$\hat{\theta}_n(X_1, \dots, X_n)$ can be chosen equal to T_n and LLN guarantees *consistency* $\hat{\theta}_n \xrightarrow{P_{\theta_0}} \theta_0$. Then, of course,

$$\sigma_n(\hat{\theta}_n) - \theta \xrightarrow{P_{\theta_0}} N_k \text{ multivariate normal distribution}$$

if $E[f_i(X)f_j(X)] < \infty$. If $\theta = g(E_\theta f(X))$ and g is differentiable, then the delta rule

$$\sqrt{n}(g(T_n) - \theta_0) \xrightarrow{P_{\theta_0}} N_k(\quad).$$

If we have an *i.i.d.* sample $X_1, X_2, \dots \stackrel{i.i.d.}{\sim} P$ and we want to test if there is evidence enough in the data to assume $P = P_0$ for some $P_0 \in M'(\mathbb{R}) = \{\text{all probability distribution on } \mathbb{R}\}$

$$H_0 : P \in M'(\mathbb{R}) \setminus \{P_0\} \quad (P \neq P_0), \quad H_1 : P = P_0.$$

Yes, such a test exists but not in a constructive way

$$H_0 : P = P_0, \quad H_1 : P \neq P_0.$$

Simplify as follows: we have a sample $\Psi_1, \dots, \Psi_n \in \mathbb{R}$ where $\Psi \sim P$ density P . Define a partition $\alpha : A_1, \dots, A_k$ of $\mathbb{R}$ and then define

$$X_1 = \sum_{j=1}^n \mathbb{I}_{\{\Psi_j \in A_1\}}, \dots, X_k = \sum_{j=1}^n \mathbb{I}_{\{\Psi_j \in A_k\}}$$

and consider $(X_1, \dots, X_k)$

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}_{\{\Psi_i \in A_l\}} \xrightarrow{a.s.} P(\Psi \in A_l)$$

as $n \rightarrow \infty$. The vector $\mathbf{X}_n = (X_1, \dots, X_k)$ has a multi-normal distribution for n trials and k bins

$$P(X_1 = x_1, \dots, X_k = x_k) = \binom{n}{x_1! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k},$$

and $x_1 + \dots + x_k = n$.

$$p_l = \int_A p(x) dx$$

$$(p_1, \dots, p_k) = a.$$

Imagine we have in mind a distribution P_0 , with density p_0 , and we want to test

$$H_0 : p = 1, \quad H_1 : p \neq a.$$

THEOREM 2.1.

If the vectors X_n are multinomially distributed with parameters n and $a = (a_1, \dots, a_k) > 0$

Proof.

□

THEOREM 2.2.

If the vectors X_n are multinomially distributed with parameters n and $p_{ij} = a_i b_j > 0$

| Exercises

EXERCISE 2.1. Suppose that the random variables X and Y are uncorrelated and have variance 1. Find the covariance matrix of the vector $(X - Y, X + Y)^T$.

EXERCISE 2.2. If the random vectors X and Y are independent, then $\text{Cov}(X+Y) = \text{Cov}X + \text{Cov}Y$. Prove this.

Delta-Method

SECTION 3.1

| Main Result and Examples

We have a model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, $\Theta \in \mathbb{R}^k$ and $\phi : \Theta \rightarrow \mathbb{R}^m$. Assume we have a sequence of estimators (T_n) for θ , but we are interested in $\phi(\theta)$, and the asymptotic properties of (T_n) going to be reflected in $\phi(T_n)$. If θ is a point where ϕ is continuous, then $T_n \xrightarrow{P_0} \theta$ implies that $\phi(T_n) \xrightarrow{P_0} \phi(\theta)$. What if $\sqrt{n}(T_n - \theta) \rightsquigarrow L_\theta$?

Taylor expansion

$$\phi(\theta + \delta_n) - \phi(\theta) = \phi'(\theta)\delta_n + o(\|\delta_n\|)$$

suggests that

$$\phi(T_n) - \phi(\theta) \approx \phi'(\theta)(T_n - \theta).$$

DEFINITION 3.1.

A function $\phi : \Theta \in \mathbb{R}^k \rightarrow \mathbb{R}^m$ is *differentiable* in an interval point $\theta \in \Theta$ if there exists a linear map $\phi'_\theta : \mathbb{R}^k \rightarrow \mathbb{R}^m$ such that

$$\phi(\theta + h) - \phi(\theta) = \phi'_\theta(h) + o(\|h\|)$$

as $h \rightarrow 0$. If $\theta \rightarrow \phi'_\theta$ is continuous, then ϕ is *continuous differentiable* at θ . The matrix ϕ'_θ is called the *total derivative* has its (i, j) -th entry

$$(\phi'_\theta)_{ij} = \frac{\partial \phi_i}{\partial \theta_j}(\theta), \quad 1 \leq i \leq m, \quad 1 \leq j \leq k.$$

LEMMA 3.2.

If, for all $1 \leq i \leq m, 1 \leq j \leq k$, $\frac{\partial \phi_i}{\partial \theta_j}$ exists in a neighbourhood of θ , and is continuous at θ , then ϕ is differentiable at θ .

THEOREM 3.3.

Let $\phi : \Theta \in \mathbb{R}^k \rightarrow \mathbb{R}^m$ be differentiable at θ . Let (T_n) be estimators for θ such that, for some (r_n) and random variable T ,

$$r_n(T_n - \theta) \rightsquigarrow T_\theta, \quad \forall \theta \in \Theta.$$

Then

$$r_n(\phi(T_n) - \phi(\theta)) \rightsquigarrow \phi'_\theta(T),$$

because

$$|r_n(\phi(T_n) - \phi(\theta)) - \phi'_\theta(r_n(T_n - \theta))| \xrightarrow{P_0} 0.$$

Proof. Since $r_n \rightarrow \infty$, $\frac{1}{r_n} \rightarrow 0$, assume $r_n(T_n - \theta) \xrightarrow{\theta} T$. □

EXAMPLE 3.4 (Sample Variance). We are often interested in estimation of EX from an *i.i.d.* sample of $X_1, X_2, \dots$, but we also want to estimate $Var(X)$, and we would like to have control over the asymptotic of our estimator (in the same way the CLT gives control over the asymptotic of estimating EX):

$$S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \bar{X}_n^2 - (\bar{X}_n)^2 = \phi(\bar{X}_n, \bar{X}_n^2).$$

$$\phi(x, y) = y - x^2.$$

Assume that

$$\alpha_1 = EX, \quad \alpha_2 = EX^2, \quad \alpha_3 = EX^3, \quad \alpha_4 = EX^4$$

all finite.

SECTION 3.2

| Variance Stabilising Transformations

Given an estimator (T_n) with limit law

$$\sqrt{n}(T_n - \theta) \xrightarrow{P_0} \mathcal{N}(0, \sigma^2(\theta)).$$

We can write that the interval

$$\left[T_n - \xi_{\frac{\alpha}{2}} \frac{\sigma(\theta)}{\sqrt{n}}, T_n + \xi_{\frac{\alpha}{2}} \frac{\sigma(\theta)}{\sqrt{n}} \right]$$

contains θ going to $1 - \alpha$. This is *not* a confidence interval, because it depends on the unknown value of θ . We may use an estimator $(\hat{\sigma}_n)$ for $\sigma(\theta)$ to obtain a valid confidence interval.

Another way to do this is by a variance stabilising transform. Assume that $\theta \in \Theta \subset \mathbb{R}$, Θ open and that $0 < \sigma(\theta) < \infty$, define

$$\phi(\theta) = \int^\theta \frac{1}{\sigma(\theta')} d\theta'$$

(the primitive of $\sigma(\theta)^{-1}$), and note that

$$\frac{d\phi}{d\theta}(\theta) = \frac{1}{\sigma(\theta)}.$$

If we now use $\phi(T_n)$ to estimate $\phi(\theta)$,

$$\sqrt{n}(\phi(T_n) - \phi(\theta)) \xrightarrow{P_0} \mathcal{N}(0, (\phi'(\theta))^2 \sigma^2(\theta)) = \mathcal{N}(0, 1).$$

Note that the right-hand side does not depend on θ any more, and

$$P\left(\phi(\theta) \in \left[\phi(T_n) - \frac{\xi_{\frac{\alpha}{2}}}{\sqrt{n}}, \phi(T_n) + \frac{\xi_{\frac{\alpha}{2}}}{\sqrt{n}}\right]\right) \rightarrow 1 - \alpha.$$

Also note that $\phi'(\theta) = \sigma(\theta)^{-1} > 0 \Rightarrow \phi(\theta)$ is strictly increasing $\Rightarrow \phi$ invertible, and the inverse ϕ^{-1} exists on $\phi(\Theta)$.

$$\begin{aligned} P_\theta^n \left(\phi(T_n) - \frac{\xi_{\frac{\alpha}{2}}}{\sqrt{n}} \leq \phi(\theta) \leq \phi(T_n) + \frac{\xi_{\frac{\alpha}{2}}}{\sqrt{n}} \right) &\rightarrow 1 - \alpha \\ \Rightarrow P_\theta^n \left(\phi^{-1} \left(\phi(T_n) - \frac{\xi_{\frac{\alpha}{2}}}{\sqrt{n}} \right) \leq \theta \leq \phi^{-1} \left(\phi(T_n) + \frac{\xi_{\frac{\alpha}{2}}}{\sqrt{n}} \right) \right) &\rightarrow 1 - \alpha. \end{aligned}$$

EXAMPLE 3.5. Let $(X, Y) \sim \rho$, and assume we have *i.i.d.* pairs (X_i, Y_i) for $i \geq 1$. Assume that $E|X^a Y^b| < \infty$ for all $0 \leq a, b \leq 4$. We wish to estimate the correlation coefficient for (X, Y) .

SECTION 3.3

| Moment Estimators

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, $\Theta \in \mathbb{R}^k$ open. Assume that for $X \sim P_\theta$, $E_\theta|X|^l < \infty$, for all $1 \leq l \leq m$ and consider the vector containing the first m moments.

$$\begin{pmatrix} \frac{1}{n} \sum_{i=1}^n X_i \\ \frac{1}{n} \sum_{i=1}^n X_i^2 \\ \vdots \\ \frac{1}{n} \sum_{i=1}^n X_i^m \end{pmatrix} \xrightarrow{P_\theta\text{-a.s.}} \begin{pmatrix} E_\theta X \\ E_\theta X^2 \\ \vdots \\ E_\theta X^m \end{pmatrix} = \phi(\theta).$$

Assume that ϕ is invertible, there exists a $\phi^{-1} : \phi(\Theta) \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$. As an estimator for θ , we then take

$$\phi^{-1} \begin{pmatrix} \bar{X}_n \\ \bar{X}_n^2 \\ \vdots \\ \bar{X}_n^m \end{pmatrix} = \hat{\theta}_n$$

to estimate θ . The Δ -rule requires that this inverse be differentiable.

We shall generalise and define a function $l : \Theta \rightarrow \mathbb{R}^m$ with

$$e(\theta) = \begin{pmatrix} E_\theta f_1(X) \\ E_\theta f_2(X) \\ \vdots \\ E_\theta f_m(X) \end{pmatrix}$$

where we assume that $E_\theta|f_i(X)f_j(X)| < \infty$ for all $1 \leq i, j \leq m$. Note that in that case,

$$\sqrt{n} \left(\begin{pmatrix} \bar{f}_{1,n} \\ \bar{f}_{2,n} \\ \vdots \\ \bar{f}_{m,n} \end{pmatrix} - e(\theta) \right) \xrightarrow{P_0} \mathcal{N}_m(0, \Sigma_\theta),$$

where

$$\bar{f}_{l,n} = \frac{1}{n} \sum_{i=1}^n f_l(X_i).$$

We want to define the moment estimator as

$$\sqrt{n} \left(e^{-1} \begin{pmatrix} \bar{f}_{1,n} \\ \bar{f}_{2,n} \\ \vdots \\ \bar{f}_{m,n} \end{pmatrix} - e^{-1}(e(\theta)) \right) \xrightarrow{P_0} \mathcal{N}_m \left(0, (e^{-1})' \Sigma_\theta (e^{-1})'^T \right).$$

THEOREM 3.6 (Moment Estimation).

Let $\Theta \in \mathbb{R}^k$ be open, suppose that $E(\theta) = E_\theta f(X)$ (with f mapping to $\mathbb{R}^m$) is one-to-one and continuously differentiable at $\theta_0 \in \Theta$ with a non-singular e'_{θ_0} . Also assume that $E_{\theta_0} \|f(x)\|^2 < \infty$.

Then $\hat{\theta}_n$ exists with probability tending to one as $n \rightarrow \infty$, and

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \overset{P_{\theta_0}}{\rightsquigarrow} \mathcal{N}\left(0, \left(e'_{\theta_0}\right)^{-1} \Sigma_{\theta_0} \left(\left(e'_{\theta_0}\right)^{-1}\right)^T\right)$$

where the matrix Σ_{θ_0} is the covariance of the vector f 's under P_{θ_0} .

Proof.

□

| Exercises

EXERCISE 3.1. Let $\hat{\lambda}_n$ be the maximum likelihood estimator of λ based on a random sample $X_1, \dots, X_n$ from the exponential distribution with parameter λ .

- (i) Find the limit distribution of the sequence $\sqrt{n}(\hat{\lambda}_n - \lambda)$.
- (ii) Construct an asymptotic confidence interval based on this.

EXERCISE 3.2. Find the limit distribution of $\sqrt{n}(\hat{\theta}_n - \theta)$ for $\hat{\theta}_n$ the maximum likelihood estimator of $\theta = P(X_i \leq 10)$ based on a random sample $X_1, \dots, X_n$ from the $\mathcal{N}(\mu, \sigma^2)$ -distribution,

- (i) when σ^2 is known.
- (ii) when both μ and σ^2 are unknown.

EXERCISE 3.3. Find the joint limit distribution of $(\sqrt{n}(\bar{X} - \mu), \sqrt{n}(S^2 - \sigma^2))$ if $\bar{X}$ and S^2 are based on a sample of size n from a distribution with finite fourth moment. Under what condition on the underlying distribution are $\sqrt{n}(\bar{X} - \mu)$ and $\sqrt{n}(S^2 - \sigma^2)$ asymptotically independent?

EXERCISE 3.4. Let $X_1, \dots, X_n$ be a sample from the 1-dimensional *exponential family* with density

$$p_{\theta}(x) = c(\theta)h(x)e^{\theta t(x)}.$$

Here h and t are known functions. The *natural parameter set* is the set

$$\Theta = \left\{ \theta \in \mathbb{R} : \int h(x)e^{\theta t(x)} dx < \infty \right\}.$$

(If X is discrete, replace the integral by a sum.)

- (i) Show that Θ is an interval.
- (ii) Show that the solution $\hat{\theta}_n$ to the likelihood equation is a moment estimator. (Remember that a score function $\dot{\ell}_{\theta} = \frac{\partial}{\partial \theta} \log p_{\theta}$ satisfies $E_{\theta} \dot{\ell}_{\theta}(X_i) = 0$.)
- (iii) Show that $\frac{\partial}{\partial \theta} E_{\theta} t(X) = \text{var}_{\theta} t(X)$ and hence is strictly positive unless $t(X)$ is degenerate.
- (iv) Show that $\sqrt{n}(\hat{\theta}_n - \theta)$ is asymptotically normal.

Z- and M-Estimators

Suppose that we have a *i.i.d.* sample $X_1, X_2, \dots$ from some P_θ , $\theta \in \Theta$. The goal is the estimation of θ . Let $m_\theta : \mathcal{X} \rightarrow \bar{\mathbb{R}}$ (assuming $X_i \in \mathcal{X}$) be defined for every $\theta \in \Theta$. Consider

$$M_n : \Theta \rightarrow \bar{\mathbb{R}} : \theta \rightarrow \frac{1}{n} \sum_{i=1}^n m_\theta(X_i) = \mathbb{P}_n m_\theta$$

a random (delta-dependent) function on Θ . The *M-estimator* associated with m_θ is the¹ value $\hat{\theta}_n(X_1, \dots, X_n)$ such that

$$M_n(\hat{\theta}_n) = \sup_{\theta \in \Theta} \{M_n(\theta) : \theta \in \Theta\}.$$

EXAMPLE 4.1 (Maximum Likelihood Estimator). Assume that there exists a finite measure μ on $\mathcal{X}$ such that $\forall \theta \in \Theta : P_\theta \ll \mu$. By the *Radon–Nikodym theorem*, there exists function $P_\theta : \mathcal{X} \rightarrow \mathbb{R}$ such that

$$P_\theta(A) = \mathbb{1}_A P_\theta(X) d\mu(X).$$

That means we can view the model as a family $\{P_\theta : \theta \in \Theta\}$ of *Radon–Nikodym density*. Choose $m_\theta(x) = \log p_\theta(x)$, then

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log p_\theta(x) = \frac{1}{n} \log \left(\prod_{i=1}^n p_\theta(x) \right).$$

The likelihood appears in $\prod_{i=1}^n p_\theta(x)$, so maximising M_n *a.e.* Θ amount to maximising the likelihood $\hat{\theta}_n$ is the *maximum likelihood estimation*.

Remark.

Going back to the general situation of *M-estimation*, we may solve

$$\Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n \psi_\theta(X_i) \tag{4.1}$$

where $\psi_\theta(x) = \frac{d}{d\theta} m_\theta(x)$. These are called *estimating equations*. If we transform (4.1) (without a underlying estimation problem) by choosing some function $\psi_\theta : \mathcal{X} \rightarrow \mathbb{R}^m$, then solving the equation $(4.1) = 0$ is called *Z-estimation*.

EXAMPLE 4.2 (Location Estimator). Let $X_1, \dots, X_n$ be a *i.i.d.* random sample of real-valued observations. Suppose we want to estimate the location of their distribution P .

¹ Subject to the condition of existence and uniqueness of such $\hat{\theta}_n$.

EXAMPLE 4.3 (Huber Estimator). If we want to robustness of the sample median and the sensitives m the bulk of the average, the we write

$$\psi(x) = \begin{cases} k, & x > k \\ x, & -h \leq x \leq h \\ -k, & x < -h \end{cases}.$$

Asymptotically, the Huber estimator is not going to the problematic (even if $P|x| = 0$), because

$$E_P|\psi_\theta(x)| = E_P|\psi(X - \theta)| \leq k.$$

SECTION 4.1

| Consistency

An estimator sequence $\hat{\theta}_n(X^n)$ is called *consistent* if $\hat{\theta}_n \xrightarrow{P_\theta} \theta$ for all $\theta \in \Theta$. The consistency of an M -estimator $\hat{\theta}_n$ depends on the asymptotic behaviour of the functions M_n in the limit. Suppose that $m_\theta : \mathcal{X} \rightarrow \bar{\mathbb{R}}$ satisfies $P_{\theta_0}|m_\theta| < \infty$ for all $\theta \in \Theta$. Then

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n m_\theta(X_i) \xrightarrow{a.s.} E_{\theta_i} m_\theta(X) = M(\theta), \quad \forall \theta \in \Theta,$$

where

$$M(\theta) = E_{\theta_i}(\log_{P_\theta}(X)).$$

THEOREM 4.4.

Let $M_n : \Theta \rightarrow \mathbb{R}$ be random functions and let $M : \Theta \rightarrow \mathbb{R}$ be a fixed function, such that for some $\theta_0 \in \Theta$:

$$\sup_{\theta \in \Theta} |M_n(\theta) - M(\theta)| \xrightarrow{P} 0, \quad (4.2)$$

$$\sup_{\theta \in \Theta: d(\theta, \theta_0) \geq \varepsilon} M(\theta) < M(\theta_0). \quad (4.3)$$

Then any sequence of estimators $\hat{\theta}_n$ that nearly maximise M_n converges in probability to θ_0 .

Remark. $X_1, X_2, \dots \stackrel{i.i.d.}{\sim} P$, and P may not lie in the model.

Proof. □

THEOREM 4.5.

Let $\Psi_n : \Theta \rightarrow \mathbb{R}^m$ be random functions and let $\Psi : \Theta \rightarrow \mathbb{R}^m$ be a fixed function, such that:

$$\sup_{\theta \in \Theta} \|\Psi_n(\theta) - \Psi(\theta)\| \xrightarrow{P} 0, \quad (4.4)$$

$$\inf_{\theta \in \Theta: d(\theta, \theta_0) \geq \varepsilon} \|\Psi(\theta)\| > \|\Psi(\theta_0)\|. \quad (4.5)$$

Then any sequence of estimators $\hat{\theta}_n$ such that $\Psi_n(\hat{\theta}_n) = o_P(1)$ converges in probability to θ_0 .

Proof. □

A class $\mathcal{F}$ of measure $f : \mathcal{X} \rightarrow \mathbb{R}$ in a *Glivenko-Cantelli class* under P , if

$$\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n f(X_i) - E_P f(X) \right| = o_P(1).$$

LEMMA 4.6.

Let (Θ, d) be a compact metric, and let $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ be a class of measurable functions, such that $\theta \mapsto f_\theta(x)$ is continuous for every $x \in \mathcal{X}$. Suppose also that there exists a function $F : \mathcal{X} \rightarrow \mathbb{R}$ such that

$$\sup_{\theta \in \Theta} |f_\theta(x)| \leq F(x), \quad \forall x \in \mathcal{X}.$$

Then

$$\sup_{\theta \in \Theta} \left| \frac{1}{n} \sum_{i=1}^n f_\theta(X_i) - E_P f_\theta(X) \right| \xrightarrow{\mathcal{P}} 0.$$

EXAMPLE 4.7 (Cauchy Likelihood). We observe $X_1, X_2, \dots$ i.i.d. from a Cauchy distribution located at $\theta \in K \subset \mathbb{R}$, K compact.

LEMMA 4.8.

Let Θ be a subset of the real line, let Ψ_n be random functions, and let Ψ be a fixed function of θ such that $\Psi_n(\theta) \xrightarrow{P} \Psi(\theta)$. Assume that the each map $\theta \mapsto \Psi_n(\theta)$ is non-decreasing, and that $\Psi_n(\hat{\theta}_n) = o_P(1)$. Let θ_0 be a point such that

$$\Psi(\theta_0 - \varepsilon) < 0 < \Psi(\theta_0 + \varepsilon), \quad \forall \varepsilon > 0.$$

Then any sequence $\hat{\theta}_n$ of near-zeroes of the Ψ_n converges to θ_0 in probability.

Proof. □

SECTION 4.2

| Asymptotic Normality

Given a consistent estimation for a finite-dimensional parameter θ , the next level of detail concerns the limit law:

$$\sqrt{n}(\theta_n - \theta) \xrightarrow{P_\theta} \mathcal{N}(0, \sigma(\theta)^2).$$

The dependence of the limiting variance σ^2 on θ represents the fact that some values of θ are harder or easier to estimate than others.

THEOREM 4.9.

Assume that the map $\theta \mapsto \psi_\theta(x)$ is twice continuously differentiable with derivatives $\dot{\psi}_\theta(x)$, $\ddot{\psi}_\theta(x)$. Assume there exists a P -integrable function $\ddot{\psi}$ with $P\ddot{\psi} < \infty$ such that,

$$\sup_{\theta \in \Theta} |\ddot{\psi}_\theta(x)| \leq \ddot{\psi}(x), \quad \forall x$$

in a neighbourhood B of θ_0 . Furthermore, suppose that

$$P\psi_{\theta_0}^2 < \infty, \quad P|\dot{\psi}_{\theta_0}| < \infty, \quad P\dot{\psi}_{\theta_0} \neq 0.$$

If $\hat{\theta}_n$ are zeroes of

$$\Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n \psi_\theta(X_i)$$

that converges in probability to a zero θ_0 of $\Psi(\theta) = P\psi_\theta$, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{P} \mathcal{N}(0, \sigma(\theta_0)^2)$$

with variance

$$\sigma(\theta_0)^2 = \frac{P\psi_{\theta_0}^2}{(P\dot{\psi}_{\theta_0})^2}.$$

Proof.

□

In the case where $\theta \in \Theta \subset \mathbb{R}^k$ and we have k estimating equations where $\Psi_n : \mathbb{R}^k \rightarrow \mathbb{R}^k$ with a (total) derivative $\dot{\Psi}_n(\theta_0)$ ($k \times k$ matrix) converges to $P\dot{\psi}_{\theta_0}$, which we are assume to be invertible. Then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{P} \mathcal{N}_k\left(0, (P\dot{\psi}_{\theta_0})^{-1} P(\psi_{\theta_0} \psi_{\theta_0}^T) (P\dot{\psi}_{\theta_0})^{-1}\right).$$

EXAMPLE 4.10 (Cauchy Likelihood). The maximum likelihood estimator for the location of a Cauchy distribution $\hat{\theta}_n$ is a zero of $\Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n \psi_\theta(X_i)$.

$$\psi_\theta(x) = \frac{x - \theta}{1 + (x - \theta)^2} = \psi(x - \theta).$$

THEOREM 4.11.

Let $\Theta \subset \mathbb{R}^k$ be open and let $x \mapsto \psi_\theta$ be a vector-valued measurable function such that for every θ_1 and θ_2 in a neighbourhood B of θ_0 , and a measurable function $\dot{\psi}(x)$ such that,

$$\|\psi_{\theta_1}(x) - \psi_{\theta_2}(x)\| \leq \dot{\psi}(x) \|\theta_1 - \theta_2\|, \quad \forall x$$

with $P\dot{\psi}^2 < \infty$. Assume that $P\|\psi_{\theta_0}\|^2 < \infty$ and that the map $\theta \mapsto P\psi_\theta$ is differentiable at a zero θ_0 , with non-singular total derivative matrix V_{θ_0} . If

$$\Psi_n(\hat{\theta}_n) = \Psi_n(\theta_0) + o_P(1)$$

near-zero of Ψ_n , and $\hat{\theta}_n \xrightarrow{\mathcal{P}} \theta_0$, then

$$\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{\mathcal{P}} \mathcal{N}_k\left(0, V_{\theta_0}^{-1}P(\psi_{\theta_0}\psi_{\theta_0}^T)(V_{\theta_0}^{-1})^T\right).$$

If $\theta \mapsto \psi_\theta(x)$ is continuous differentiable in a neighbourhood of θ with derivative $\dot{\psi}_\theta(x)$, then

$$\begin{aligned} \|\psi_{\theta_1}(x) - \psi_{\theta_2}(x)\| &= \left\| \int_0^1 \dot{\psi}_{\theta_1+t(\theta_2-\theta_1)}(x) dt (\theta_2 - \theta_1) \right\| \\ &\leq \sup_{0 \leq t \leq 1} \|\dot{\psi}_{\theta_1+t(\theta_2-\theta_1)}(x)\| \|\theta_2 - \theta_1\| \\ &\leq \sup_{\theta \in B} \|\dot{\psi}_\theta(x)\| \|\theta_2 - \theta_1\| \\ &= \dot{\psi}(x) \|\theta_2 - \theta_1\|. \end{aligned}$$

EXAMPLE 4.12 (Huber Estimator).

Suppose we have two estimators $\hat{\theta}_{1,n}$ and $\hat{\theta}_{2,n}$, both of which are asymptotic normal:

$$\begin{aligned} \sqrt{n}(\hat{\theta}_{1,n} - \theta) &\xrightarrow{\mathcal{P}} \mathcal{N}(0, \sigma_1^2(\theta)), \\ \sqrt{n}(\hat{\theta}_{2,n} - \theta) &\xrightarrow{\mathcal{P}} \mathcal{N}(0, \sigma_2^2(\theta)). \end{aligned}$$

We prefer $\hat{\theta}_{1,n}$ over $\hat{\theta}_{2,n}$ if $\sigma_1^2(\theta) < \sigma_2^2(\theta)$. And we define *asymptotic relative efficiency* as

$$\left[\hat{\theta}_{1,n} - \xi_{\frac{\alpha}{2}} \frac{\sigma_1(\theta)}{\sqrt{n}}, \hat{\theta}_{1,n} + \xi_{\frac{\alpha}{2}} \frac{\sigma_1(\theta)}{\sqrt{n}} \right].$$

SECTION 4.3

| Maximum Likelihood Estimators

Consider a sample $X_1, X_2, \dots \stackrel{i.i.d.}{\sim} P_{\theta_0}$ for some $\theta_0 \in \Theta$. We assume that the model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is dominated by some σ -finite measure μ .² Write $P_\theta(X)$ for the R-N derivative $\frac{dP_\theta}{d\mu} : \mathbb{R} \rightarrow [0, \infty)$. We need these densities to write down

$$M_n(\theta) = \frac{1}{n} \sum_{i=1}^n \log \left(\frac{P_\theta(X_i)}{P_{\theta_0}(X_i)} \right) \rightarrow \hat{\theta}_n(X_1, \dots, X_n),$$

$$M(\theta) = E_{\theta_0} \log \left(\frac{P_\theta(X)}{P_{\theta_0}(X)} \right).$$

M is called the *Kullback-Leibler divergence*. The maximum likelihood estimator $\hat{\theta}_n(X_1, \dots, X_n)$ is a (near)-maximises of M .

DEFINITION 4.13.

A parametrisation $\theta \mapsto P_\theta$ is *identifiable* if it is injective ($\theta_1 \neq \theta_2 \Rightarrow P_{\theta_1} \neq P_{\theta_2}$).

Remark.

LEMMA 4.14.

If $\mathcal{P}$ has an identifiable parametrisation, then $\theta \mapsto M(\theta)$ has a unique maximisation in θ_0 .

Proof.

□

DEFINITION 4.15.

The *Hellinger distance* is

$$H : \mathcal{P} \times \mathcal{P} \rightarrow [0, \infty),$$

$$H(P, Q) = \left(\int \left(\sqrt{p(x)} - \sqrt{q(x)} \right)^2 d\mu(x) \right)^{1/2}, \quad \mu = P + Q.$$

To prove consistency of the maximum likelihood estimator $\hat{\theta}_n$, we suppose that

$$\sup\{|M_n(\theta) - M(\theta)| : \theta \in \Theta\} \xrightarrow{P} 0.$$

Alternatively, we may switch to Z-estimator with some factors

$$\Psi_n(\theta) = \frac{1}{n} \sum_{i=1}^n \frac{\dot{P}_\theta(X_i)}{P_\theta(X_i)}, \quad \psi_\theta(X) =$$

So once consistency is established, what about asymptotic normality? Note the form of asymptotic covariance:

$$(P_{\theta_0} \ddot{\ell}_{\theta_0})^{-1} P_{\theta_0} \left(\dot{\ell}_{\theta_0} \dot{\ell}_{\theta_0}^T \right) \left(P_{\theta_0} \ddot{\ell}_{\theta_0}^T \right)^{-1}, \quad (4.6)$$

$$V_{\theta_0}^{-1} P_{\theta_0} \left(\dot{\ell}_{\theta_0} \dot{\ell}_{\theta_0}^T \right) \left(V_{\theta_0}^T \right)^{-1}, \quad V_{\theta_0} = \frac{\partial}{\partial \theta} P_{\theta_0} \dot{\ell}_\theta |_{\theta=\theta_0}. \quad (4.7)$$

² For example, Lebesgue measure for $X \in \mathbb{R}$, σ -counting measure $\#$ if $X \in \mathbb{Z}$.

EXAMPLE 4.16.

THEOREM 4.17.

| Exercises

EXERCISE 4.1. Let $p_{\mu, \sigma^2}(x)$ be the density of the $\mathcal{N}(\mu, \sigma^2)$ -distribution and let P an arbitrary probability distribution (not necessarily normal).

- (i) Calculate $M(\mu, \sigma^2) = P \log p_{\mu, \sigma^2}$
- (ii) For which (μ, σ^2) does $M(\mu, \sigma^2)$ attain its maximal value?
- (iii) What does this suggest about the behaviour of the estimators $(\hat{\mu}_n, \hat{\sigma}_n^2)$ that maximise $(\mu, \sigma^2) \rightarrow \mathbb{P}_n \log p_{\mu, \sigma^2}$? Is this true?

EXERCISE 4.2. Define

$$\psi(x) = \begin{cases} 1-p, & x < 0 \\ 0, & x = 0 \\ -p, & x > 0 \end{cases}.$$

Show by an informal argument that, under appropriate conditions, the corresponding Z-estimator is asymptotically normal with zero mean and asymptotic variance

$$\frac{p(1-p)}{f(F^{-1}(p))^2}.$$

EXERCISE 4.3. Let $X_1, \dots, X_n$ be independent $\mathcal{N}(\mu, 1)$ -distributed random variables. Define $\hat{\theta}_n$ as the point of minimum of the function

$$\theta \rightarrow \sum_{i=1}^n (X_i - \theta)^4.$$

- (i) Show that $\hat{\theta}_n \xrightarrow{P} \theta_0$ for some θ_0 . Which θ_0 ?
- (ii) Show that $\sqrt{n}(\hat{\theta}_n - \theta_0)$ converges to a normal distribution.
- (iii) Determine the asymptotic relative efficiency of $\hat{\theta}_n$ and the sample mean as an estimator of μ .

EXERCISE 4.4. Calculate the Kullback-Leibler divergence between two exponential distributions with different scale parameters. When is it maximal?

EXERCISE 4.5. Calculate the Kullback-Leibler divergence between two normal distributions with different location and scale parameters. When is it maximal?

EXERCISE 4.6. Calculate the Kullback-Leibler divergence between two Laplace distributions with different locations. When is it maximal?

EXERCISE 4.7. Let $X_1, \dots, X_n$ be *i.i.d.* $\mathcal{N}(\mu, \sigma^2)$ -distributed.

- (i) Calculate the Fisher information I_{μ, σ^2} in one observation for the parameter (μ, σ^2) and its inverse;
- (ii) Verify by direct calculations that the maximum likelihood estimator for (μ, σ^2) is asymptotically normal with mean zero and covariance matrix I_{μ, σ^2}^{-1} .

References

- [1] A. W. van der Vaart. *Asymptotic Statistics*, volume 3 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge, 1998.